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Mapping data practices in gaming across publishing platforms, community sites, and online multiplayer games

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What are common data practices amongst gaming platforms?

Motivation

Digital platforms increasingly collect and process larger amounts of data to enhance the experience they offer. However, trust in these practices is not always given. Scholarship on such privacy matters across social media platforms is continuously growing. Yet the same is not the case for the gaming context.

We examined data practices across three types of gaming platforms to gain a more holistic view on the conditions that shape the online experiences of gamers. Our objective is to formulate a foundational understanding for the types of data collected in gaming context and toto bridge debates between social media scholarship and games research.

Cases

Publishing: Apple App Store, Epic Store, Google Play Store, Nintendo, PlayStation, Riot Games, Roblox, Steam, WeGame, Xbox.

Games: Apex Legends, Brawl Stars, Call of Duty, Clash of Clans, Counter-Strike, DotA2, FIFA, Fortnite, League of Legends, Minecraft, Mobile Legends, PUBG, Roblox, Rocket League, Valorant.

Community: Discord, Douyu, FACEIT, Facebook, Huya, KOOK, PerfectWorld Arena, Reddit, Twitch, YY, YouTube Gaming.

Method

Privacy policies as core body of data

Open coding of initial cases by each author

Development of four key thematic groups

Coding of all cases and secondary data
(e.g. blog posts, news articles, etc.)

Development of themes + list of practices

Consent, Control, and Security

Automatic collection: platforms collect a variety of data automatically, such as device information, web tracking, voice and text chats, or types of usage data, including all kinds of in-game actions.

Opt-in collection: active opt-in from the user is mostly required for more sensitive data categories such as personal, payment, health, or geolocation related. Same for underage user account management.

User control: platforms gave users the standard options to check, edit, and delete data (including accounts). Some noted possibility of privacy tools and browser settings to limited collection of data. Options varied based on regional level.

Security: next to basics, platforms inform users that 100% security is impossible and shift some responsibility on them.

Marketing and Business Operations

Marketing and Advertising: platforms use a variety of user data (identifiers, personal, usage, gameplay, etc) to run their own marketing and advertising campaigns. This includes email marketing, promotions, events, but also on-platform activations.

Customer Support: platforms outliend how they use variety of personal and technical data in order to manage custom relations and resolve support requests.

Research and Analytics: we found that all platforms outlined the use of data for analytics purposes. This ranged from improving specific features towards more general development of the platforms.

Management and Legal: platforms process data within the scope of cooperation with law enforcement or to run their business (e.g. financial reporting requirements).

Platform Experience

Personalization: platforms use data to personalize experiences. This overlapped with other social media in terms of localization, curation of content, but also specific elements such as matchmaking (e.g. personalizing who a person can play with).

Moderation: we found a mix of general community moderation and monitoring systems. But also gaming specific anti-cheat systems, accessing devices on the kernel-level (e.g. core of operating system), standing out from other social media.

Technical Maintenance: platforms and game specifically collect lots of technical data for maintenance and system security.

Platform access: platforms outlined that in order to access services, users have to share data, making data an access condition and foundation for features like matchmaking.

Third-party Operations

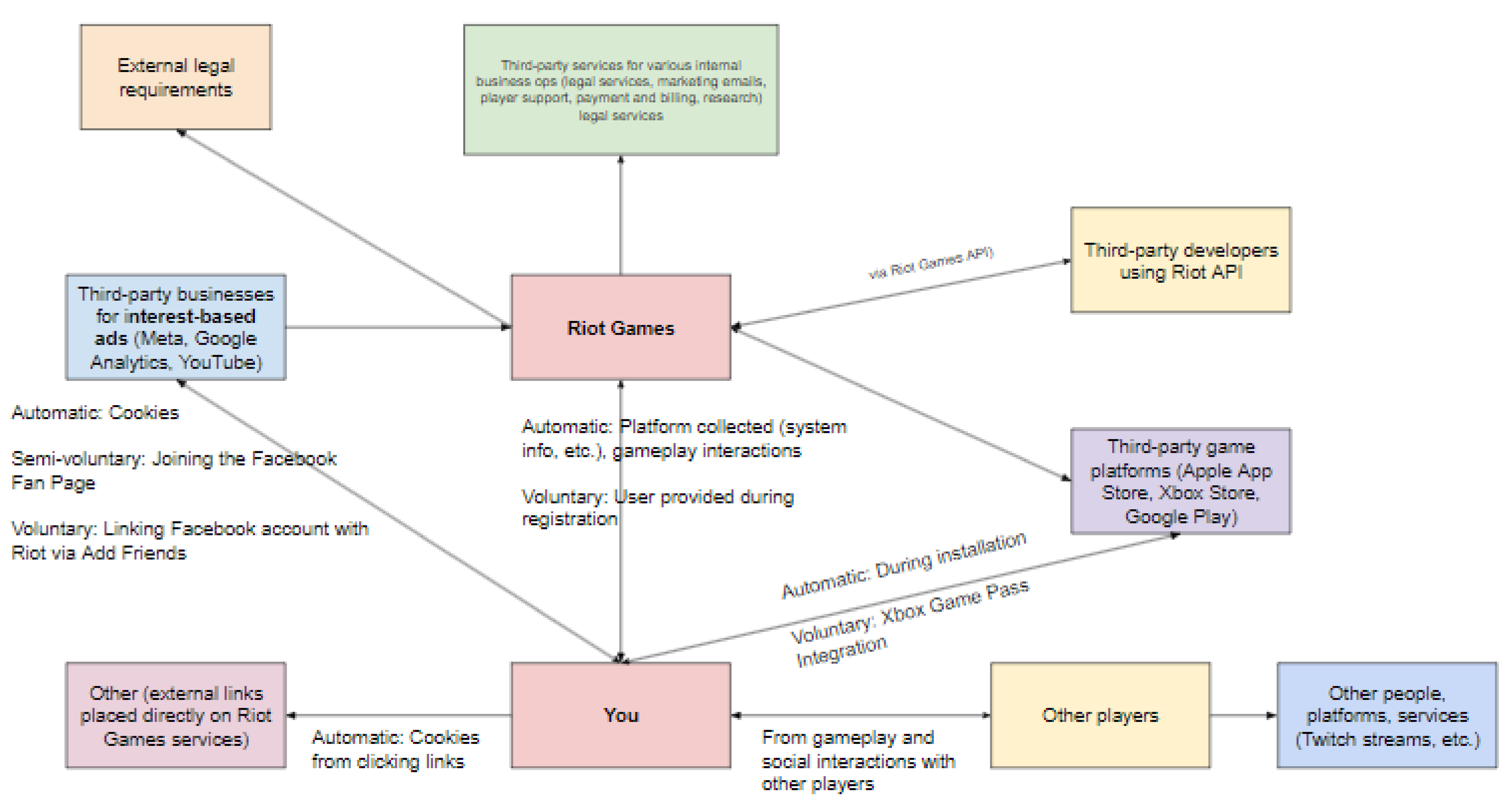
Sharing data: all platforms noted sharing data with third-parties, although not all were selling it. Most common was sharing in advertising networks. However, also more functional like for payment processing. Also, gaming-specific in terms of APIs for video games and stats tracking sites, for example.

Receiving data: almost all platforms received data from third parties, mostly tracking and audience measurement related segmentation data to enrich own sources.

Collecting data: platforms allow certain partners to collect data on their sites. Either through their cookies, for instance, or games hosted on their platform.

Service providers: the most common relationship was advertisers. However, also analytics partners, game developers, developers of bots or add-on features.

Visualisation of data practices and relationships for the case of Riot Games (League of Legends)



Future Work and Limitations

🚧 **Privacy policies only offer a limited insight into what platforms actually do.** We found that often the language in policies remained vague. Secondary documents allowed to add some depth. Overall, a limitation thus remains and we're looking for ways to address further in the project.

🔍 **Further exploration of platforms and additional, technical sources.** For example: around consent management and opt-ins. Also, unique gaming practices, like anti-cheat, which access people's computing devices at significant depth (e.g. kernel) unlike regular social media sites.

💭 **Conduct qualitative research with gamers** to understand how they experience data practices. Furthermore, understand how they negotiate a sense of trust in these, also in comparison to their use of other social media sites to gain deeper comparative insights.

📊 **Conduct quantitative research based on findings** to check levels of awareness and literacy when it comes to the use of data in the context of gaming. Aim is to outline common areas of concern as well as help better educate gamers about data.